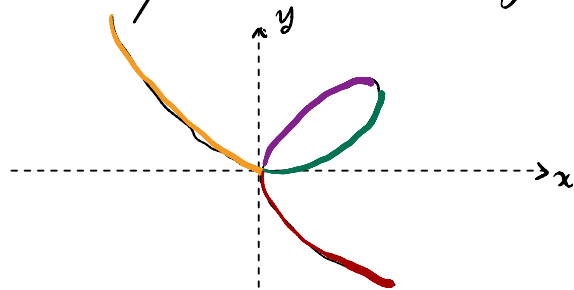


Lecture 9, Jan 9th, 2025.

For example let $x^3 + y^3 = 6xy$.



Implicit Differentiation. we don't need to solve an equation for y in terms of x in order to find the derivative of y .

Instead we can use implicit differentiation. This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for $\frac{dy}{dx}$.

Example. a). Find y' if $x^3 + y^3 = 6xy$.

b). find the tangent line at $(3,3)$.

c). At what point in the first quadrant is the tangent line horizontal.

Sln. a).

$$\frac{d}{dx}(x^3 + y^3) = \frac{d}{dx}(6xy)$$

$$\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) = 6 \frac{d}{dx}(xy)$$

$$3x^2 + 3y^2 y' = 6 \left(\frac{d}{dx}(x) \cdot y + x \cdot y' \right)$$

$$3x^2 + 3y^2 y' = 6(y + xy')$$

$$\text{Solve for } y': \quad 3x^2 + 3y^2 y' = 6y + 6xy'$$

$$3x^2 - 6y = 6xy' - 3y^2 y'$$

$$3x^2 - 6y = y'(6x - 3y^2)$$

$$y' = \frac{3x^2 - 6y}{6x - 3y^2} = \frac{x^2 - 2y}{2x - y^2}.$$

b). at $(3, 3)$
 x, y .

$$m = y' \Big|_{(3,3)} = \frac{3^2 - 2(3)}{2(3) - 3^2} = -1.$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -1(x - 3)$$

$$y = -x + 3 + 3 = -x + 6.$$

c). we are looking for $y' = 0 \Leftrightarrow \frac{x^2 - 2y}{2x - y^2} = 0 \Leftrightarrow x^2 = 2y \Leftrightarrow y = \frac{1}{2}x^2$.

we know that, $x^3 + y^3 = 6xy$.

$$x^3 + \left(\frac{1}{2}x^2\right)^3 = 6x\left(\frac{1}{2}x^2\right)$$

$$x^3 + \frac{1}{8}x^6 = 3x^3$$

$$x^6 = 16x^3$$

$$x^3(x^3 - 16) = 0$$

$$x = 0 \quad \text{or} \quad x^3 = 16 \quad x = \sqrt[3]{16} = 2^{4/3}.$$

if $x = 0$ then $y = 0$.

$$\text{if } x = 2^{4/3} \quad y = \frac{1}{2}(2^{4/3})^2 = 2^{5/3}.$$

then, the tangent line is horizontal at $(0, 0)$ and $(2^{4/3}, 2^{5/3})$.

Example. Find y' if $\sin(x+y) = y^2 \cos x$.

Sln. $\frac{d}{dx}(\sin(x+y)) = \frac{d}{dx}(y^2 \cos x)$

$$\text{By C.R.} \rightarrow \cos(x+y) \cdot \frac{d}{dx}(x+y) = \frac{d}{dx}(y^2) \cdot \cos x + y^2 \cdot \frac{d}{dx}(\cos x).$$

$$\cos(x+y)(1+y') = 2yy' \cos x - y^2 \sin x.$$

$$\left[G(y) = y^2 \quad G(y) = F(H(y)) \quad F(x) = x^2 \quad H(y) = y \quad \text{By C.R.} \quad G'(y) = 2y y' \right]$$

$$\cos(x+y) + y' \cos(x+y) = 2y y' \cos x - y^2 \sin x.$$

$$\cos(x+y) + y^2 \sin x = 2y y' \cos x - y' \cos(x+y).$$

$$\cos(x+y) + y^2 \sin x = y'(2y \cos x - \cos(x+y)).$$

$$y' = \frac{\cos(x+y) + y^2 \sin x}{2y \cos x - \cos(x+y)}.$$

Second derivatives of implicit functions.

Example. Find y'' if $x^4 + y^4 = 16$.

i). y' :

$$\frac{d}{dx}(x^4 + y^4) = \frac{d}{dx}(16).$$

$$4x^3 + 4y^3 \cdot y' = 0.$$

$$y' = -\frac{x^3}{y^3}.$$

ii). y'' :

$$\frac{d}{dx}(y') = \frac{d}{dx}\left(-\frac{x^3}{y^3}\right)$$

$$y'' = \frac{-3x^2 \cdot y^3 - (-x^3)(3y^2 \cdot y')}{(y^3)^2}.$$

$$y'' = \frac{-3x^2 y^3 + 3x^3 y^2 y'}{y^6}$$

$$y'' = \frac{-3x^2 y^3 + 3x^3 y^2 \left(-\frac{x^3}{y^3}\right)}{y^6}$$

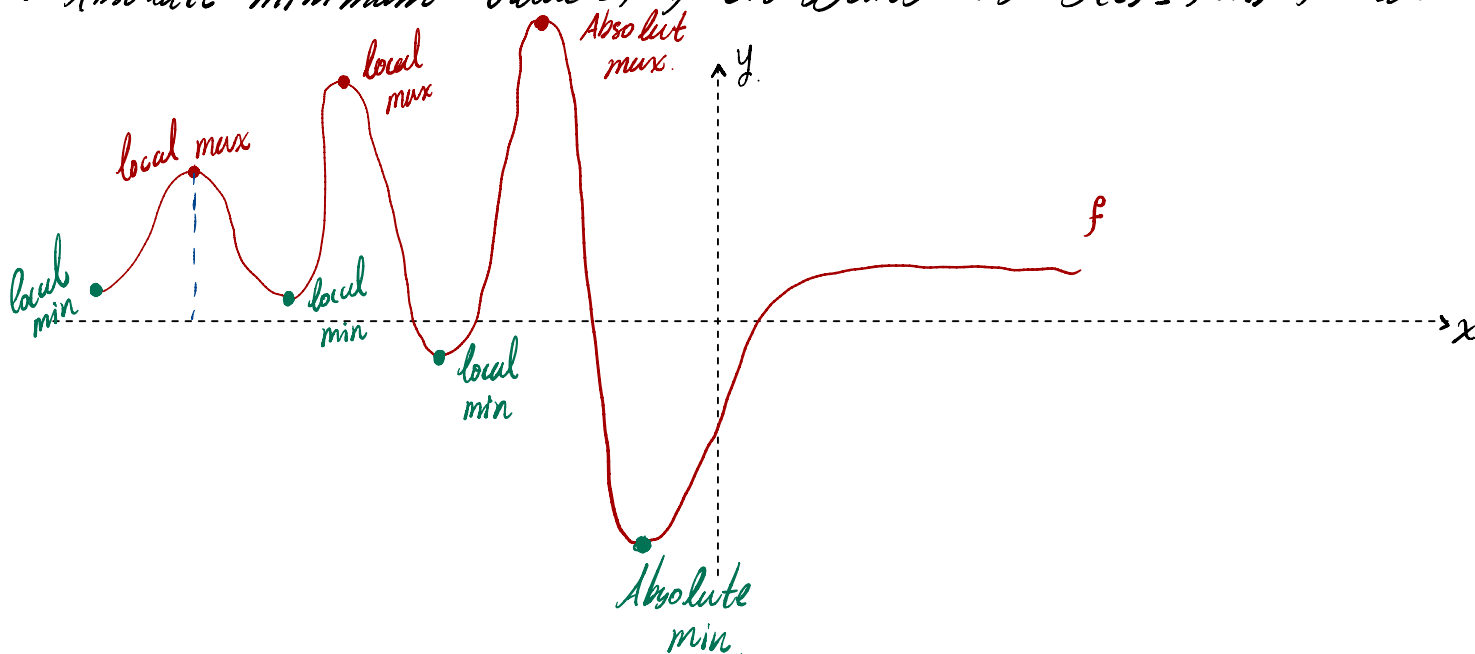
$$y'' = \frac{\frac{-3x^2 y^3}{1} - \frac{3x^6}{y}}{y^6}$$

$$y'' = \frac{-3x^2 y^4 - 3x^6}{y^7} = \frac{-3x^2(y^4 + x^4)}{y^7} = \frac{-48x^2}{y^7}.$$

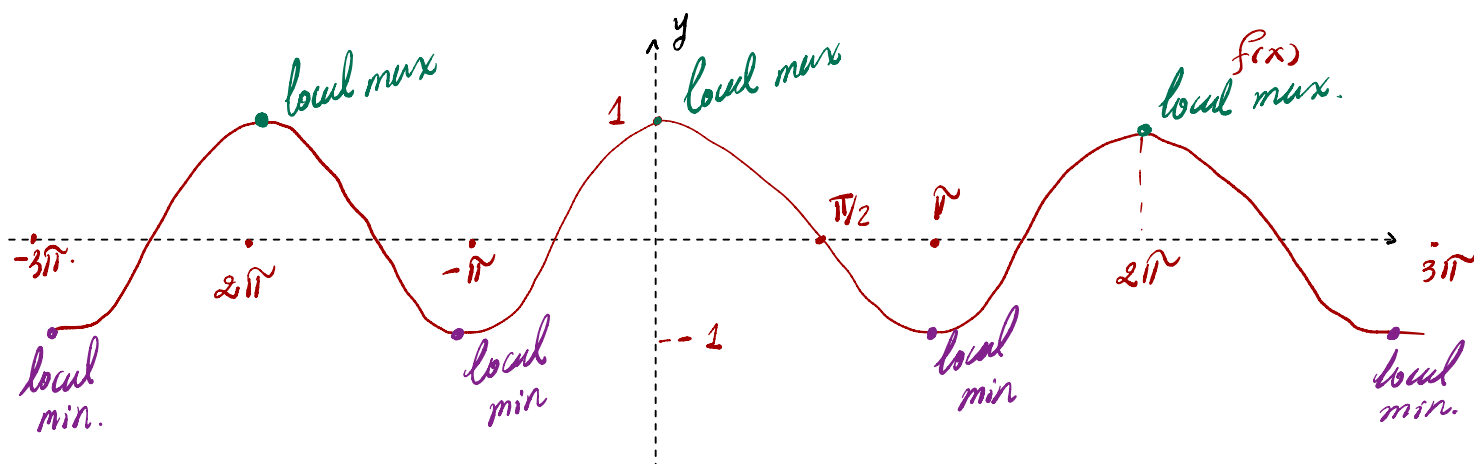
3.1. Maximum and minimum values.

Def. Let c be a number in $\text{Dom} f$. Then $f(c)$ is the:

- Absolute maximum value of f on $\text{Dom} f$ if $f(c) \geq f(x)$ for all $x \in \text{Dom}$.
- Absolute minimum value of f on $\text{Dom} f$ if $f(c) \leq f(x)$ for all $x \in \text{Dom}$.



Example. Let consider $f(x) = \cos x$, this function on $\mathbb{R}$ has only local minimum and local maximum.



Thm. (The Extrem Value theorem) if f a continuous function on a closed interval $[a, b]$ then f attains an absolute maximum $f(c)$ and an absolute minimum $f(d)$ for some points $c, d \in [a, b]$.

Critical Numbers and closed interval Method.

Let f is a function then the values where $f'(x) = 0$ for $x \in \text{Dom} f$.

are called critical numbers of f .

Now, in particular $\text{Dom } f = [a, b]$ then the critical values of f are the values s.t. $f'(x) = 0$ and $x = a$, $x = b$.

Thm (Fermat's theorem). If f has a local maximum or minimum at c , and $f'(c)$ exists, then $f'(c) = 0$.

(+). (New critical number). We say that the function f has a critical value at c if $f'(c) = 0$ or $f'(c)$ does not exist.

Example. Find the critical points of.

a). $f(x) = x^3 - 3x^2 + 1$

b). $f(x) = x^{3/5}(4-x)$

Sol. a). $\text{Dom } f = \mathbb{R}$. $f'(x) = 3x^2 - 6x$. Since $f'(x)$ exists for $x \in \mathbb{R}$, the only critical values are $f'(x) = 0$ that is $x = 0$, $x = 2$.

b). $\text{Dom } f = \mathbb{R}$ Then, by the product rule.

$$f'(x) = \frac{3}{5} x^{3/5-1} (4-x) + x^{3/5} (-1)$$

$$= -x^{3/5} + \frac{3(4-x)}{5x^{2/5}} = \frac{-5x + 12 - 3x}{5x^{2/5}} = \frac{12-8x}{5x^{2/5}}$$

then: i). $f'(x) = 0 \Leftrightarrow 12-8x=0 \Leftrightarrow x = \frac{3}{2}$.
ii). $f'(x) \text{ DNE} \Leftrightarrow 5x^{2/5}=0 \Leftrightarrow x=0$ } are the critical points.

Prop. If f has a maximum or minimum at c , then c is critical value of f .

The closed interval method. To find the absolute maximum and minimum values of a continuous function f on a closed interval $[a, b]$.

1. Find the values of f at the critical values of f in (a, b) .
2. Find the values of f at the endpoints of the interval $[a, b]$.

3. The largest of the values from step 1 and 2 is the absolute maximum; the smallest of these values is the absolute minimum.

Example. Find the absolute maximum and minimum of

$$f(x) = x^3 - 3x^2 + 1 \quad -\frac{1}{2} \leq x \leq 4.$$

Sln. $f(x)$ is continuous on $[-\frac{1}{2}, 4]$. We can use the closed interval method we know that

$$f'(x) = 3x^2 - 6x = 3x(x-2)$$

So, the critical points are $x=0$, $x=2$. Notice that each of these lies in the interval $(-\frac{1}{2}, 4)$.

Step 1 $f(0) = 1$, $f(2) = -3$.

Step 2. $f(-\frac{1}{2}) = -\frac{1}{8} - \frac{3}{4} + 1 = \frac{1}{8}$; $f(4) = 4^3 - 3(16) + 1 = 17$.

Step 3. then $f(2) = -3$ is the smallest then is the absolute minimum.
the $f(4) = 17$ is abs. max.